MIL-V-63118A(AR)
16 February 1978

SUPERSEDING
MIL-V-63118(AR)
6 April 1977

MILITARY SPECIFICATION

VULCANIZED RUBBER OBTURATOR

This specification is approved for use by US Army Armament Research and Development Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the manufacture of a vulcanized rubber obturator to be used in the 105MM, APDS-T ammunition.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078	Ammunition, Standard Quality Assurance Provisions, General Specification for
MIL-R-3065	Rubber, Fabricated Products

FSC: 1315

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-1169 Packaging, Packing and Marking for Shipment of Inert Ammunition Components

DRAWING

ARMY

9322081 - Rubber Compound
9322080 - Obturator
9323862 - Obturator (Heavy)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D395 - Compression Set of Vulcanized Rubber
D412 - Tension Testing of Rubber
D573 - Accelerated Aging of Vulcanized Rubber by the Oven Method
D624 - Tear Resistance of Vulcanized Rubber
D792 - Specific Gravity and Density of Plastics by Displacement
D2000 - Standard Classification System for Rubber Products in Automotive Applications
D2240 - Indentation Hardness of Rubber and Plastics by Means of a Durometer
D2290 - Apparent Tensile Strength of Ring or Tubular Plastics by Split Disc Method

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103)

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with MIL-R-3065, ASTM D-2000, other applicable specifications and standards.

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3.2 Obturators. Obturators shall be manufactured from an uncured homogeneous rubber compound in the form of a continuous ring (no seam or joint) such that when cured or vulcanized in accordance with standard practices for fabricating rubber products the vulcanized product shall comply with all the requirements on drawing 9322080 or 9323862 and all applicable specifications and standards.

3.3 Physical and mechanical properties.

3.3.1 Rubber compound (test specimens). The rubber compound submitted for testing in a test speciment form shall meet the following requirements in accordance with drawing 9322080 or 9323862.

a. Specific gravity.
b. Compression set.
c. Hardness before and after heat aging.
d. Tensile strength and ultimate elongation before and after heat aging.
e. Tear resistance.
f. Low-temperature brittleness.

3.3.2 Obturators. Obturators shall meet the following requirements in accordance with drawing 9322080 or 9323862.

a. Specific gravity.
b. Load to break and extension at break before heat aging.
c. Load to break and extension at break after heat aging.

3.4 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.5 Workmanship. Obturators shall be free from surface imperfections, cracks, porosity, voids and other defects likely to affect their performance. Trimming of flash must be accomplished in a manner which will not destroy or damage the surface features of the trimmed part.

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4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspection. The following types of inspection shall be conducted on this item:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample, as designated by the Contracting Officer, for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the quantities stipulated in TABLE I which have been manufactured using the same production processes, procedures, molds, dies and equipment which will be used in fulfilling the contract. All materials shall be obtained from the same sources of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract, and submit a record of this inspection with the sample, including statements of findings for materials, processes and tests. A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer. A first article sample, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing, specification or source of supply as to significantly affect product uniformity as determined by the Government. After first article submission, any new molds introduced in the process must be qualified, i.e., the contractor must submit a first article sample in accordance with Section II of TABLE I for each mold introduced in the process.

4.3.2 Inspection to be performed. See MIL-A-48078 and TABLE I specified herein.

4.3.3. Rejection. See MIL-A-48078.

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TABLE I. FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-V-63118A(AR)

PARAGRAPH CATEGORY	TITLE / EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	DRAWING NUMBER / NEXT HIGHER ASSEMBLY / PARAGRAPH REFERENCE /INSPECTION METHOD
	OBTURATOR		SHEET 1 OF 1		9322080 or 9323862
	I RUBBER COMPOUND (TEST SPECIMENS)				
	A. Physical and Mechanical Properties				
	1. Specific gravity	10		3.3.1	4.5.1.1
	2. Compression set	10		3.3.1	4.5.1.2
	3. Hardness before and after heat aging	10		3.3.1	4.5.1.3
	4. Tensile, strength and ultimate elongation before and after heat aging	20		3.3.1	4.5.1.4
	5. Tear resistance	10		3.3.1	4.5.1.5
	6. Low-temperature brittleness	10		3.3.1	4.5.1.6
	II OBTURATORS				
	A. Examination for defects	Notes	1&2	3.2	4.4.2.1 and 4.4.2.2
	B. Physical and mechanical properties				
	1. Specific gravity	Notes	1&2	3.3.2	4.5.2.1
	2. Load to break and extension at break before heat aging	Notes	1&2	3.3.2	4.5.2.2
	3. Load to break and extension at break after heat aging	Note	1	3.3.2	4.5.2.3
	4. Hardness before and after heat aging.	5/mold	(Info only)		4.5.2.4

NOTES: 1. Used one obturator per cavity and a minimum of twenty obturators per mold.
2. After doing II-A, the same samples can be used to do II-B-1 and then II-B-2.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.1.1 Obturator lot. The inspection obturator lot shall be composed of no more than 5,000 obturators which were made from one lot of rubber compound (see 6.3).

4.4.1.2 Rubber compound lot. Each ingredient used to make the rubber compound lot shall comply with the following:

a. Natural rubber from not more than one source of supply.

b. Zinc oxide from not more than one lot from one manufacturer.

c. Stearic acid from not more than one lot from one manufacturer.

d. Pine tar (if required) from not more than one lot from one source of supply.

e. Paraffin wax from not more than one lot from one manufacturer.

f. Carbon black from not more than one lot from one manufacturer. If more than one type of carbon black is used, each type shall be from not more than one lot from one manufacturer.

g. N-phenyl-2-naphthylamine from one manufacturer.

h. Mercapto-benzthiazole from one manufacturer.

i. Sulfur from one manufacturer.

j. Retarder (if required) from one manufacturer.

4.4.2 Examination. See MIL-A-48078.

Sampling plans. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105 Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-V-63118A(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.1	OBTURATOR				9322080
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Material defective		.40%	3.5	Visual
102	Rear face cut, torn or damaged		.40%	3.5	Visual/Manual
103	Thickness of aft lip		.40%	3.2	Gage
104	Diameter outside at rear		.40%	3.2	Gage
Minor					
201	Diameter, inside small		.65%	3.2	Gage
202	Diameter outside at front		.65%	3.2	Gage
203	Thickness through web		.65%	3.2	Gage
204	Length from rear of web to rear face		.65%	3.2	Gage
205	Length total		.65%	3.2	Gage
206	Marking missing, incorrect or illegible		.65%	3.2	Visual

NOTES

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-V-63118A(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.2.	OBTURATOR (HEAVY)				9323862
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Material defective		.40%	3.5	Visual
102	Rear face cut, torn or damaged		.40%	3.5	Visual/Manual
103	Thickness of aft lip		.40%	3.2	Gage
104	Diameter outside at rear		.40%	3.2	Gage
Minor					
201	Diameter, inside small		.65%	3.2	Gage
202	Diameter, inside at rear		.65%	3.2	Gage
203	Diameter, outside at front		.65%	3.2	Gage
204	Thickness through web		.65%	3.2	Gage
205	Length from rear of web to rear face		.65%	3.2	Gage
206	Length of flat at rear		.65%	3.2	Gage
207	Length total		.65%	3.2	Gage
208	Marking missing, incorrect or illegible		.65%	3.2	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-V-63118A(AR)

PARAGRAPH	TITLE			SHEET 1 OF 1	DRAWING NUMBER
4.4.2.3	OBTURATOR (PHYSICAL AND MECHANICAL PROPERTY TESTS)				9322080 or 9323862
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
	Rubber Compound (Test Specimens)				
101	Specific gravity	10		3.3.1	4.5.1.1
102	Compression set	10		3.3.1	4.5.1.2
103	Hardness before and after heat aging	10		3.3.1	4.5.1.3
104	Tensile strength and ultimate elongation before and afrer heat aging	20		3.3.1	4.5.1.4
105	Tear resistance	10		3.3.1	4.5.1.5
106	Low-temperature brittleness	10		3.3.1	4.5.1.6
	Obturators				
107	Specific gravity	Note A		3.3.2	4.5.2.1
108	Load to break and extension at break before heat aging	Note A		3.3.2	4.5.2.2
109	Load to break and extension at break after heat aging	Note B		3.3.2	4.5.2.3

NOTES: A. See 4.4.3.2.1 and 4.4.3.2.2
B. See 4.4.3.2.3

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4.4.3 Testing.

4.4.3.1 Rubber Compound (Test Specimens). Physical and mechanical properties - Major defect.

Test specimens must be submitted for testing (see 6.5):

a. Just before starting the manufacturing of the first obturator lot.

b. For every 20,000 (or portion thereof) obturators manufactured that used the same lot of rubber compound.

c. Every time a new rubber compound lot is used in the manufacturing of obturators.

4.4.3.1.1 Specific gravity test. Ten test specimens shall be sumitted to the specific gravity test. If any sample fails to meet the specific gravity requirement as specified in drawing 9322080 or 9323862, the rubber compound lot shall be rejected. See 4.5.1.1.

4.4.3.1.2 Compression set test. Ten test specimens (can use the same test specimens after completion of 4.4.3.1.1) shall be submitted to the compression set test. If any sample fails to meet the compression set requirement in accordance with drawing 9322080 or 9323862 the rubber compound lot shall be rejected. See 4.5.1.2.

4.4.3.1.3 Hardness test before and after heat aging. The hardness shall be measured in ten test specimens. Then the test specimens shall be heat aged, and the hardness shall be measured again. If any sample fails to meet the hardness requirement before and after heat aging in accordance with drawing 9322080 or 9323862, the rubber compound lot shall be rejected. See 4.5.1.3.

4.4.3.1.4 Tensile strength and ultimate elongation tests before and after heat aging. The tensile strength and ultimate elongation shall be measured in ten test specimens. Another ten test specimens shall be heat-aged and the tensile strength and ultimate elongation shall be measured again. If any sample fails to meet the tensile strength and ultimate elongation requirement before and after heat aging in accordance with drawing 9322080 or 9323862, the rubber compound lot shall be rejected. See 4.5.1.4.

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4.4.3.1.5 Tear resistance test. The tear resistance shall be measured in ten test specimens. If any sample fails to meet the tear resistance requirement in accordance with drawing 9322080 or 9323862, the rubber compound lot shall be rejected. See 4.5.1.5.

4.4.3.1.6 Low-temperature brittleness test. Ten test specimens shall be prepared and submitted to the low-temperature brittleness test. Five test specimens shall be tested in accordance with procedure A and the other five in accordance with procedure B. If any test specimen when tested in accordance with 4.5.1.6 shows fractures or cracks, this shall be cause for rejection of the rubber compound lot.

4.4.3.2 Obturators. Physical and mechanical properties - Major defect.

For the purpose of selecting sample obturators for the physical and mechanical properties testing, the homogeneous obturator lot shall not contain more than 5000 units.

Beginning with the first lot produced and continuing until three consecutive lots have complied with the applicable requirements specified, a sample of 80 obturators shall be randomly selected from each lot for the tests listed in paragraphs 4.4.3.2.1, 4.4.3.2.2 and 4.4.3.2.3.

After three consecutive lots have met the criteria of paragraphs 4.4.3.2.1, 4.4.3.2.2 and 4.4.3.2.3, the sample size shall be reduced to 50 obturators randomly selected from each lot which will be submitted for the tests listed in paragraph 4.4.3.2.1, 4.4.3.2.2 and 4.4.3.2.3.

4.4.3.2.1 Specific gravity test. Half of the sample obturators selected in accordance with 4.4.3.2 shall be submitted to the specific gravity test. If any obturator sample fails to meet the specific gravity requirement in accordance with drawing 9322080 or 9323862, the obturator lot shall be rejected. See 4.5.2.1.

4.4.3.2.2 Load to break and extension at break test before heat aging. After completion of 4.4.3.2.1, the same obturator samples shall be submitted to the load to break and extension at break test. If any obturator sample fails to meet the load to break and extension at break requirement in accordance with drawing 9322080 or 9323862, the obturator lot shall be rejected. See 4.5.2.2.

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4.4.3.2.3 Load to break and extension at break test after heat aging. Half of the obturators selected in accordance with 4.4.3.2 shall be heat aged and then submitted to the load to break and extension at break test. If any obturator sample fails to meet the load to break and extension at break requirements after heat aging in accordance with drawing 9322080 or 9323862, the obturator lot shall be rejected. See 4.5.2.3.

4.4.3.2.4 Hardness before and after heat aging. (Info only.) Hardness data shall be obtained from five of the samples used in 4.4.3.2.3 for information purposes. The hardness measurements shall be recorded before heat aging. Then after heat aging, the hardness measurements of the same five samples shall be recorded again. The hardness information data shall be obtained in such a fashion to relate to the 4.4.3.2.3 data. Hardness, load to break and extension at break data shall be provided to the contracting officer for forwarding to ARRADCOM, DRDAR-LCR-Q, Dover, N.J. 07801, for informational purposes.

4.4.4. Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 MIL-A-48078.

4.5 Method of inspection.

4.5.1 Rubber compound (test specimens).

4.5.1.1 Specific gravity test. The test specimens shall conform in shape to that specified in Method B, Type 1, ASTM D395. The specific gravity test shall be performed in accordance with ASTM D792, Method A-1.

4.5.1.2 Compression set test. The compression set test shall be performed in accordance with ASTM D395, Method B. The test specimens shall have the dimensions as specified in ASTM D395, Method B, Type 1.

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4.5.1.3 Hardness test before and after heat aging. The test specimens shall conform in shape to that specified in Method B, Type 1, ASTM D395. The hardness test shall be performed in accordance with ASTM D2240, use a Type A durometer. The heat aging shall be accomplished in accordance with ASTM D573.

4.5.1.4 Tensile strength and ultimate elongation test before and after heat aging. The test specimens shall conform in shape to die "c" ASTM D412. The tensile strength and ultimate elongation test shall be performed in accordance with ASTM D412. The heat aging shall be accomplished in accordance with ASTM D573.

4.5.1.5 Tear resistance test. The test specimens shall conform in shape to die "c" ASTM D624. The tear resistance test shall be performed in accordance with ASTM D624. The test shall be made at 23 ± 2°C (73.4 ± 3.6°F), also the test specimens shall be conditioned at this temperature for at least 3 hours before testing.

4.5.1.6 Low-temperature brittleness test. The test specimens shall conform in shape to die "c" ASTM D412 and be tested according to the two procedures described herein.

a. Procedure A. The test specimens shall be clamped in a loop position such that the two opposite ends of the specimens are in intimate contact and clamped together with a suitable clamp (use a binder clip or equivalent) across the specimen at .250" + .125" from the ends. The clamped specimens shall then be placed in a cold chamber (can use dry ice) for 16 hours at -62.2 ± 5.6°C (-80 ± 10° F). The specimens shall be removed from the cold chamber and examined in the loop area for fractures or cracks. The clamps shall be removed and the specimens will be allowed to relax to their original straight flat position. The specimens shall be examined again for fractures or cracks.

b. Procedure B. The test specimens shall be placed flat in a cold chamber (can use dry ice) for 16 hours at -62.2 ± 5.6°C (-80 ± 10° F). At the end of the conditioning period, the specimens shall be removed from the cold chamber. After 5 seconds at room temperature, the test specimens shall be bent to form a loop such that their opposite ends are in intimate contact. The looped area shall be examined for fractures or cracks.

Note: When performing procedure A or B, the looped area or the area to be looped shall never be touched by the human hands.

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4.5.2 Obturators.

4.5.2.1 Specific gravity test. The specific gravity of the entire obturator shall be checked in accordance with ASTM D792, Method A-3. A balance capable of weighing to 0.01 gram shall be used.

4.5.2.2 Load to break and extension at break test before heat aging. Load to break and extension at break test shall be performed following the criteria of ASTM D2290 Procedure A with the following exceptions:

a. The split disc test fixture and accessories shall conform to figures 1 thru 3.

b. The cross-head speed shall be 2 to 5 inches/minute.

c. The split discs shall be separated until each half is in contact with the specimen without applying a load to the specimen. The distance of separation (initial separation shall then be measured. Split disc separation upon specimen rupture (ultimate spearation) shall also be measured. Inches of elongation shall be determined by subtracting initial separation from ultimate separation.

4.5.2.3 Load to break and extension at break tests after heat aging. Heat aging (70 hours at 70 ± 2° C) shall be accomplished in accordance with ASTM D573 (where applicable). See 4.5.2.2 for the performance of the load break and extension at break test.

4.5.2.4 Hardness test before and after heat aging. (Info. only). The hardness shall be checked with a durometer, Type A as specified in ASTM D2240. A minimum of three equally spaced hardness readings shall be taken in the approximate center of the outside diameter of each sample. An average hardness result shall be determined for each obturator.

5. PACKAGING.

5.1 Packaging, packing and marking. Packaging, packing and marking shall be level c, as specified in MIL-STD-1169.

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6. NOTES.

6.1 Ordering data. See MIL-A-48078.

6.2. Submission of inspection equipment designs for approval. See MIL-A-48078. Submit equipment designs, as required, to Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAT-T, Dover, NJ 07801.

6.3 Rubber compound lot. A rubber compound lot is to consist of a single mix or two or more homogeneous mixes of ingredients in accordance with drawing 9322081.

Two or more mixes are considered to belong to one rubber compound lot when each mix will be made by the same manufacturer under uniform conditions. These mixes must have the same ingredients (see 4.4.1.2) and be made to the same formulation by weight. These mixes can be made at different times and do not need to be consolidated together.

6.4 Process control.

6.4.1 Rubber compound. Once mixing of ingredients has taken place, the identity of the mix must be maintained. For every mix, the contractor must provide a certificate indicating each ingredient used with its proportion by weight. This certificate shall give full details of the actual composition used, including the proprietary name of each ingredient. Process control shall be maintained to assure adequate mixing and homogeneity of mixes.

6.4.2 Obturators. Once a satisfactory processing technique has been established in the manufacture of the obturators, no alteration may be made without the approval of the proper agency. A vulcanizing temperature of 141°C (285.8°F) is normally regarded as appropriate. Vulcanizing temperature of 160°C (320°F) and above are regarded as unsuitable. There is to be an efficient control of the time and temperature of vulcanization such as to produce a reliable and consistent product. A written record of the processing cycle (temperature, time, etc.) shall be maintained for each obturator lot manufactured. Also, the contractor must keep a record of the identification (with each ingredient used with its formulation by weight) of each mix or homogeneous mixes used in the manufacturing of each obturator lot.

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6.5 Preparation of test specimens.

The test specimens shall be prepared in accordance with the specified ASTM's (see 4.5). The test specimens shall be vulcanized at the temperature appropriate for the compound, and as used for parts fabricated from it. It is advised that the test specimens shall be tested prior to the conditions stated in 4.4.3.1, a, b and c. If the contractor uses a rubber compound material for the obturator manufacturing without testing acceptance approval and later this rubber compound material fails the acceptance test criteria, all obturators manufactured from this material shall be rejected.

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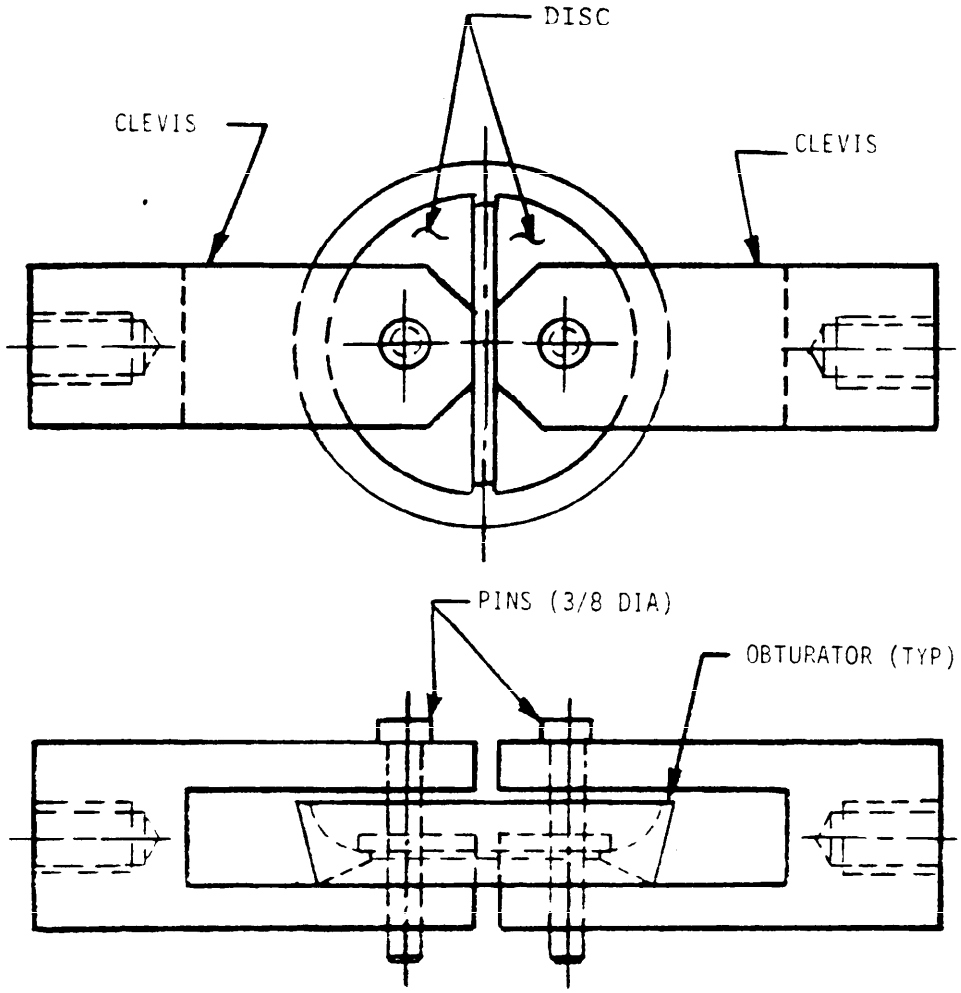


FIGURE 1. Fixture
SCALE 1/2

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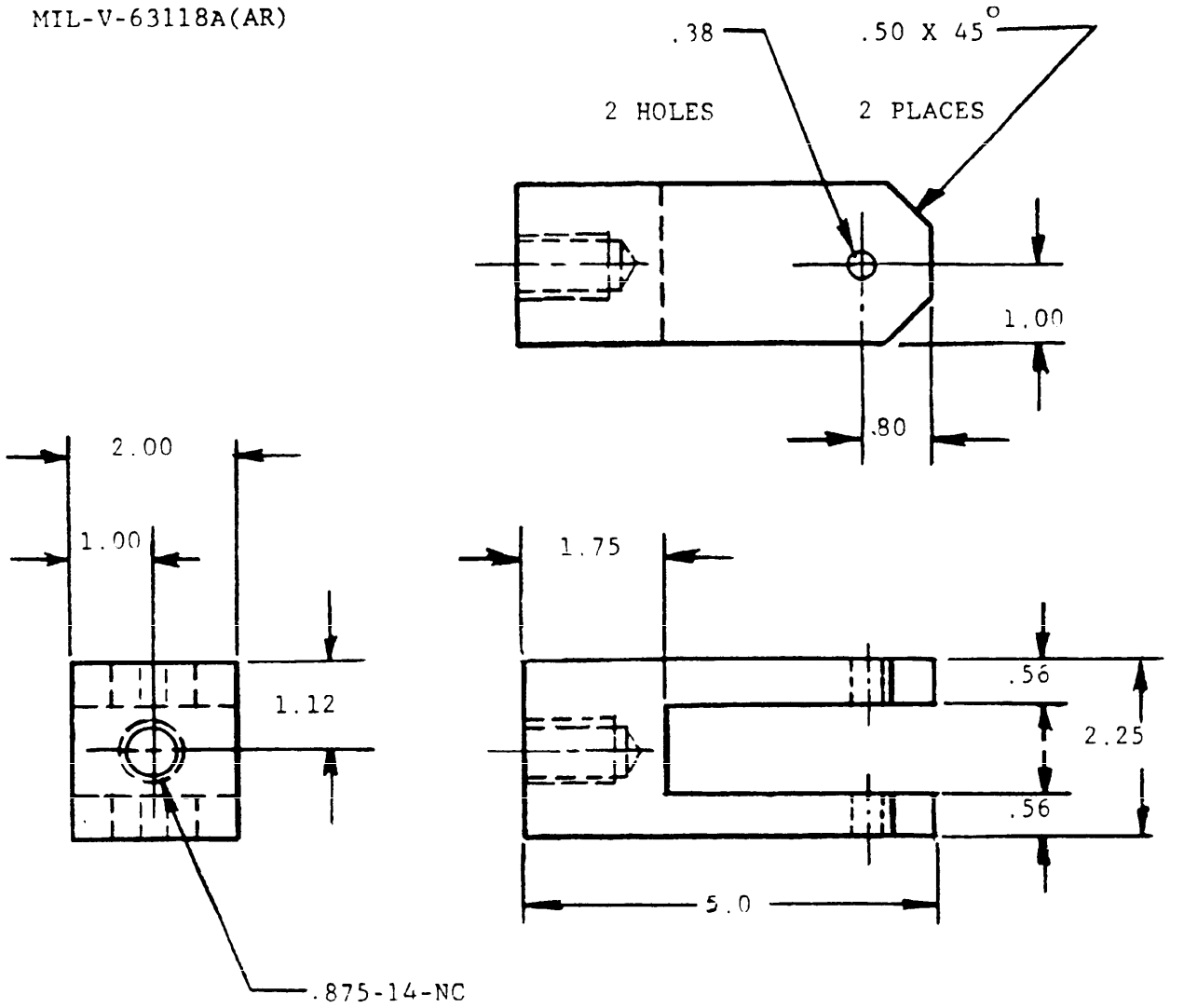


FIGURE 2. CLEVIS (2 REQUIRED)
SCALE 1/2

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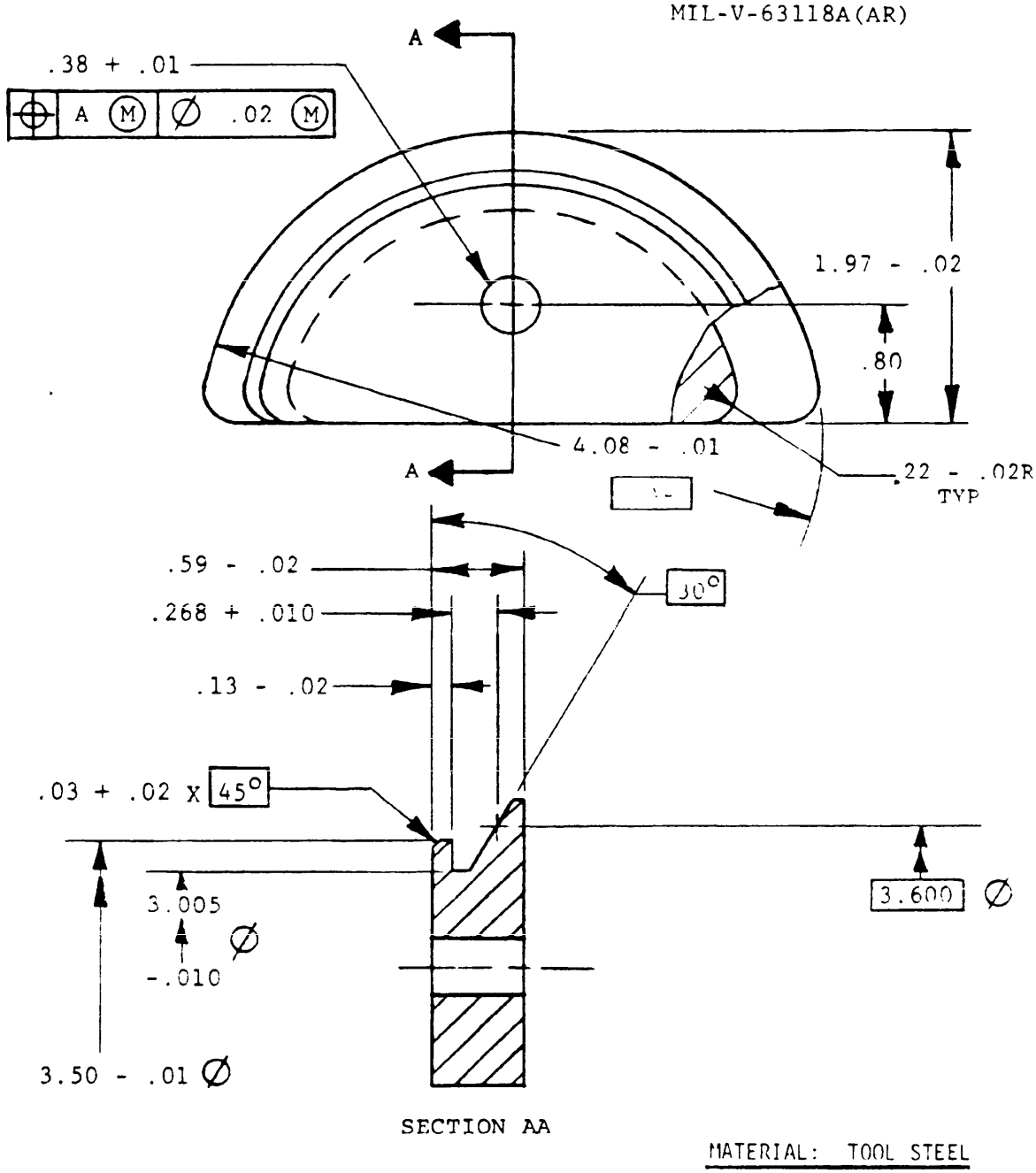


FIGURE 3. DISC (2 REQUIRED)

SCALE 1/1

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